

CHE 201 Spring 2020 Final Exam

Anna Christensen

TOTAL POINTS

100 / 100

QUESTION 1

1 15 pts

1.1 a 12 / 12

✓ - **0 pts** Correct

- **2 pts** first is incorrect
- **2 pts** second is incorrect
- **2 pts** third is incorrect
- **2 pts** fourth is incorrect
- **2 pts** fifth is incorrect
- **2 pts** sixth is incorrect

1.2 b 3 / 3

✓ - **0 pts** Correct

- **3 pts** Not done or wrong
- **2 pts** not explained

QUESTION 2

2 10 pts

2.1 a 5 / 5

✓ - **0 pts** Correct

- **1 pts** calc error
- **3 pts** h₂s is wrong
- **2 pts** formula is wrong
- **2 pts** explanation is not clear
- **5 pts** not done

2.2 b 5 / 5

✓ - **0 pts** Correct

- **2 pts** pr(T₁) is wrong
- **1 pts** calculation error
- **5 pts** not done or wrong
- **3 pts** formula is wrong

QUESTION 3

3 10 pts

3.1 a 2 / 2

✓ - **0 pts** Correct

- **2 pts** Not done or wrong

3.2 b 2 / 2

✓ - **0 pts** Correct

- **2 pts** wrong or not done
- **1 pts** one is wrong

3.3 c 6 / 6

✓ - **0 pts** Correct

- **6 pts** not done or wrong
- **2 pts** u value is wrong
- **2 pts** s value is wrong

QUESTION 4

4 32 / 32

✓ - **0 pts** Correct

- **2 pts** sketch is not drawn or incorrect
- **3 pts** engineering model is not listed
- **4 pts** simplified energy balance is not done
- **4 pts** work is not calculated or wrong
- **4 pts** heat is not calculated or wrong
- **5 pts** entropy change is not calculated
- **5 pts** sigma is not calculated
- **2 pts** pv diagram is wrong
- **3 pts** Ts diagram is wrong
- **2 pts** v₂ is wrong
- **2 pts** u₂ is wrong
- **2 pts** calculation error
- **2 pts** s₂ is wrong
- **3 pts** did not multiplied with mass
- **5 pts** v₁, u₁, s₁ are wrong
- **2 pts** T in sigma calc is wrong

QUESTION 5

5 5 33 / 33

✓ - 0 pts Correct

- 3 pts sketch is not drawn
- 3 pts engineering model is not listed
- 9 pts mass flow rate of inlet not calculated or

wrong

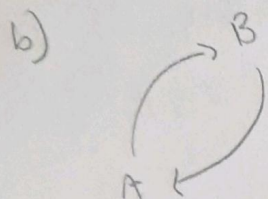
- 9 pts mass flow rate of outlet is not calculated
- 9 pts entropy production is not calculated
- 2 pts s_3 is wrong
- 2 pts h_3 is wrong
- 2 pts h_2 is wrong
- 2 pts s_2 is wrong

- ① False
 a) False
True
True
False
True

reversible: $\sigma = 0$

$$d\psi = -s dT - P dv$$

$\nearrow 0$
 $\therefore d\psi < 0 \quad dv = (+)$



The heat transfer for the irreversible pathway is less than zero.

$\Delta S_{A \rightarrow B} = 0$ because $Q = 0$ because it's adiabatic and $\sigma = 0$ because it's reversible and $\Delta S = \int \left(\frac{\delta Q}{T} \right)_b + \sigma$

$\Delta S_{\text{cycle}} = 0$ because it's a state property.

$$\Delta S_{\text{cycle}} = \Delta S_{AB} + \Delta S_{BA} \quad \text{so} \quad \Delta S_{B \rightarrow A} = 0$$

if $\Delta S_{BA} = 0$ then $0 = \frac{Q}{T} + \sigma \Rightarrow \frac{Q}{T} = -\sigma$

σ for $B \rightarrow A$ is positive because it's irreversible

so $\frac{Q}{T} < 0$ therefore $Q < 0$.

1.1 a 12 / 12

✓ - 0 pts Correct

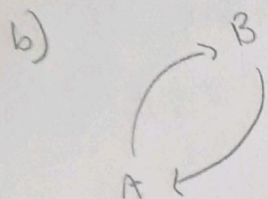
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1.2 b 3 / 3

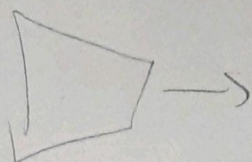
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2

a) vapor
 $P_1 = 10 \text{ bar}$
 $T_1 = 300^\circ\text{C}$



$P_2 = 5 \text{ bar}$

$W = 150 \frac{\text{kJ}}{\text{kg}}$

Table A-4 $P = 10 \text{ bar}$ $T = 300^\circ\text{C}$

$$\eta = \frac{W_{\text{cv}}/m}{h_1 - h_2}$$

$$h_1: \frac{320 - 300}{320 - 280} = \frac{3093.9 - h_1}{3093.9 - 3008.2}$$

$$h_2: \frac{7.2307 - 7.1214}{7.2307 - 7.0592} = \frac{2939.9 - h_2}{2939.9 - 2855.4}$$

$$h_1 = 3051.1 \frac{\text{kJ}}{\text{kg}}$$

$$h_2 = 2886.2 \frac{\text{kJ}}{\text{kg}}$$

$$s_1: \frac{320 - 300}{320 - 280} = \frac{7.1962 - s_1}{7.1962 - 7.0465}$$

$$s_1 = 7.1214$$

$$\eta = \frac{150 \frac{\text{kJ}}{\text{kg}}}{3051.1 - 2886.2 \frac{\text{kJ}}{\text{kg}}}$$

$$s_2 = s_1 = 7.1214 \frac{\text{kJ}}{\text{kg K}}$$

$$\eta = 0.91$$

b)

$T_1 = 650 \text{ K}$
 $P_1 = 10 \text{ bar}$

$s_2 = s_1$
 $P_2 = 2 \text{ bar}$

$T_2 = ?$

$\Pr(T_1) = 21.86$

$$\Pr(T_2) = \Pr(T_1) \left(\frac{P_2}{P_1} \right) = 21.86 \left(\frac{2 \text{ bar}}{10 \text{ bar}} \right) = 4.372 = \Pr(T_2)$$

~~420 K~~ between 420 and 410 K

$$\frac{420 - T_2}{420 - 410} = \frac{4.522 - 4.372}{4.522 - 4.153}$$

$$T_2 = 415.9 \text{ K}$$

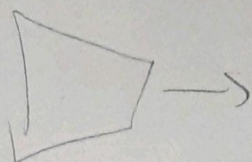
2.1 a 5 / 5

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- 3 pts h2s is wrong
- 2 pts formula is wrong
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- 5 pts not done

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③ a) $g(T, P): \quad -\left(\frac{ds}{dP}\right)_T = \left(\frac{dv}{dT}\right)_P$

$$dg = -s dT + v dP$$

b) $dh = T ds - v dP:$

$$\frac{\partial^2 h}{\partial P \partial s} = \left(\frac{\partial T}{\partial P}\right)_s$$

$$\frac{\partial^2 h}{\partial P \partial s} = \left(\frac{\partial v}{\partial s}\right)_P$$

c) 3 kg water $P = 10 \text{ bar}$ $T = 150^\circ\text{C} \rightarrow$ compressed liquid
use table A-2 same temp

$\psi = ?$ $\psi = -s dT - P dv$

$$\psi = u - Ts$$

$$u = 631.68 \frac{\text{kJ}}{\text{kg}}$$

$$s = 1.8418 \frac{\text{kJ}}{\text{kg K}}$$

$$T = 150 + 273 = 423 \text{ K}$$

$$\psi = 3 \text{ kg} \left[631.68 \frac{\text{kJ}}{\text{kg}} - (423 \text{ K}) \left(1.8418 \frac{\text{kJ}}{\text{kg K}} \right) \right]$$

$$\psi = -442 \text{ kJ}$$

3.1 a 2 / 2

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3.3 C 6 / 6

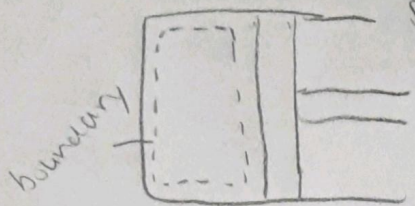
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- 2 pts u value is wrong
- 2 pts s value is wrong

4

3kg water

$$T_1 = 120^\circ\text{C} \quad P_1 = 3\text{bar} = P_2$$



25°C reservoir

engineering model

- closed system
- constant pressure
- $KE = PE = 0$
- work is in form of expansion, ΔV

$$W = ?$$

$$Q = ?$$

$$\sigma = ? \text{ - reversible / irreversible}$$

energy balance:

$$\Delta E = \Delta U + \Delta PE + \Delta KE = Q - W$$

$$\Delta U = Q - W$$

entropy balance

$$\Delta S = \int \frac{Q_{\text{total}}}{T} + \sigma$$

$$W = \int P dV = P \int dV = P[V_2 - V_1]$$

$$V_1 = 1.0603 \times 10^{-3} \frac{\text{m}^3}{\text{kg}} = P(m(V_2 - V_1))$$

$$V_2 = 0.364$$

$$W = 3\text{bar} \times \left(\frac{10^5 \frac{\text{N}}{\text{m}^2}}{\text{bar}} \right) \left(\frac{1 \text{KJ}}{1000 \text{J}} \right)$$

$$\times 3\text{kg} (0.364 - 1.0603 \times 10^{-3} \frac{\text{m}^3}{\text{kg}})$$

$$W = 326.65 \text{ kJ}$$

$$\Delta U = Q - W$$

$$m(u_2 - u_1) + W = Q$$

$$3\text{kg} (1750.6 - 503.50) + 326.65 = Q$$

$$Q = 4067.95 \text{ kJ}$$

$$\Delta S = m(s_2 - s_1) = 3\text{kg} (4.866 - 1.5276) = 10.02 \frac{\text{kJ}}{\text{K}}$$

$$\Delta S = \frac{Q}{T} + \sigma \rightarrow 10.02 \frac{\text{kJ}}{\text{K}} = \frac{4067.95 \text{ kJ}}{(25+273)\text{K}} + \sigma \Rightarrow \sigma = -3.63 \frac{\text{kJ}}{\text{kg}}$$

impossible

comp liquid
4-2

	P	T	V	S	U
State 1	3 bar	120°C	1.0603 × 10 ⁻³	1.5276	503.50
State 2 x=0.6	3 bar	133.6°C	0.364	4.866	1750.6

$$V_f = 1.0732 \times 10^{-3} \quad V_g = 0.6058$$

$$V_2 = (1-0.6)(1.0732 \times 10^{-3}) + 0.6(0.6058)$$

$$V_2 = 0.364 \frac{\text{m}^3}{\text{kg}}$$

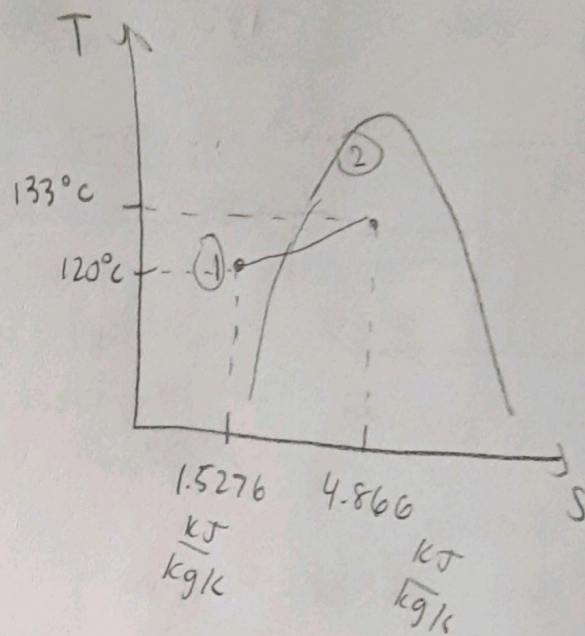
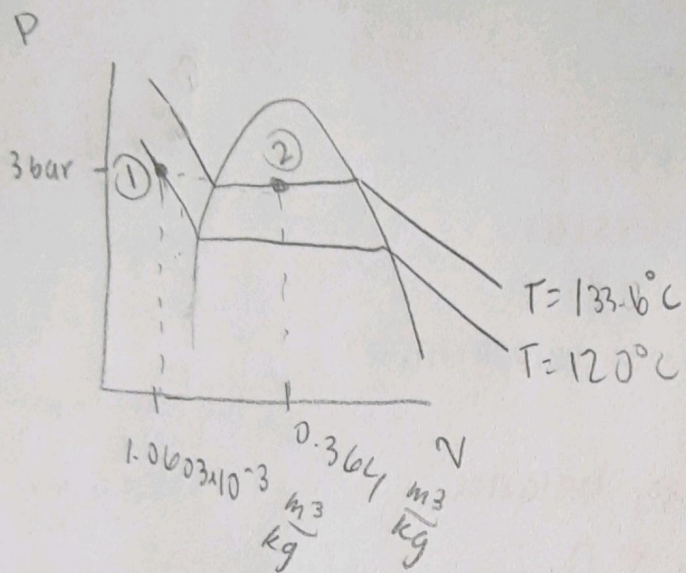
$$S_f = 1.6718 \quad S_g = 6.9919$$

$$S_2 = (0.4)(1.6718) + 0.6(6.9919) = 4.866 \frac{\text{kJ}}{\text{kgK}}$$

$$u_f = 561.15 \quad u_g = 2543.6 \frac{\text{kJ}}{\text{kg}}$$

$$u_2 = (0.4)(561.15) + 0.6(2543.6)$$

$$u_2 = 1750.6 \frac{\text{kJ}}{\text{kg}}$$

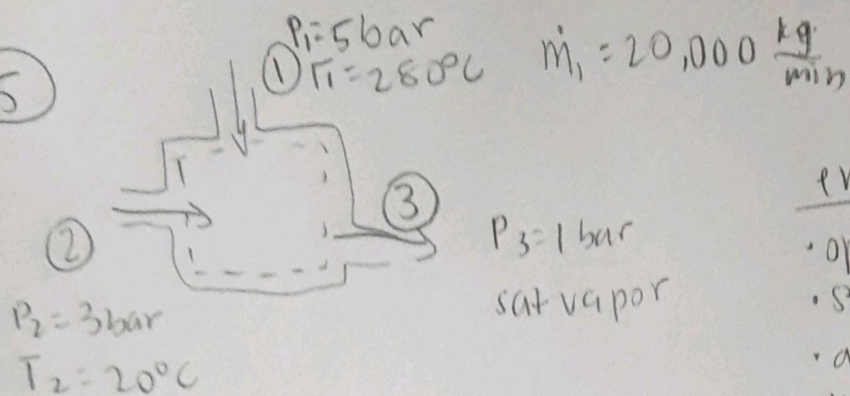


4 4 32 / 32

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- 5 pts sigma is not calculated
- 2 pts pv diagram is wrong
- 3 pts Ts diagram is wrong
- 2 pts v2 is wrong
- 2 pts u2 is wrong
- 2 pts calculation error
- 2 pts s2 is wrong
- 3 pts did not multiplied with mass
- 5 pts v1,u1,s1 are wrong
- 2 pts T in sigma calc is wrong

5



engineering model

- open system
- steady state
- adiabatic $Q_{cv} = 0$
- $KE = PE = 0$
- $W_{cv} = 0$

$$\dot{m}_2 = ?$$

$$\dot{m}_3 = ?$$

$$\dot{S} = ?$$

mass balance

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

energy balance

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i (h_i + KE_i + PE_i) - \sum \dot{m}_e (h_e + KE_e + PE_e)$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3$$

entropy balance

$$\frac{dS_{cv}}{dt} = \sum \frac{\dot{Q}_j}{T_j} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{S}_{cv}$$

$$\dot{m}_1 s_1 + \dot{m}_2 s_2 - \dot{m}_3 s_3 + \dot{S}_{cv} = 0$$

	P	T	h	S
1	5 bar	280°C	3022.9	7.3865
2	3 bar	20°C	83.96	0.2966
3	1 bar	99.63°C	2675.5	7.3594

$\frac{kJ}{kg \cdot K}$

$$\dot{m}_2 = ?$$

$$\dot{m}_3 = \dot{m}_1 + \dot{m}_2$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 - (\dot{m}_1 + \dot{m}_2) h_3 = 0$$

$$\dot{m}_1 (h_1 - h_3) + \dot{m}_2 (h_2 - h_3) = 0$$

$$\dot{m}_2 = \dot{m}_1 \left(\frac{h_3 - h_1}{h_2 - h_3} \right)$$

$$\begin{aligned} h_1 &= 3022.9 \frac{kJ}{kg} \\ h_2 &= 83.96 \frac{kJ}{kg} \\ h_3 &= 2675.5 \frac{kJ}{kg} \end{aligned}$$

$$\dot{m}_2 = 20,000 \frac{kg}{min} \times \frac{1 min}{60 s} \left(\frac{2675.5 - 3022.9}{83.96 - 2675.5} \right)$$

liquid entering

$$\dot{m}_2 = 44.68 \frac{kg}{sec}$$

vapor exiting chamber

$$\dot{m}_3 = \dot{m}_1 + \dot{m}_2 = \left(20,000 \frac{kg}{min} \times \frac{1 min}{60 s} \right) + 44.68 = 378 \frac{kg}{sec} = \dot{m}_3$$

$$\dot{S} = \dot{m}_3 s_3 - \dot{m}_1 s_1 - \dot{m}_2 s_2 = \left(378 \frac{kg}{sec} \right) (7.3594) - (20,000 \frac{kg}{min} \times \frac{1 min}{60 s}) (7.3865) - (44.68) (0.2966)$$

$$\begin{aligned} s_1 &= 7.3865 \frac{kJ}{kg \cdot K} \\ s_2 &= 0.2966 \frac{kJ}{kg \cdot K} \\ s_3 &= 7.3594 \frac{kJ}{kg \cdot K} \end{aligned}$$

$$= 2781.9 - 13.25 - 2461.9$$

$$= 306.75 \frac{kJ}{K \cdot sec} = \dot{S}$$

5 5 33 / 33

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